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COMP 2235 – Networks I

Lab Report: Ping & Traceroute Utilities

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Q1 – How Ping Works

Ping Overview

According to the *Wikipedia Article that was given,* Ping is a tool used to check whether a computer on a network is reachable. It also measures the round-trip time it takes for data to travel to the target computer and back. According to Microsoft TechNet, Ping works by sending *ICMP Echo Request* messages to the target and receiving *ICMP Echo Reply* messages in return. The round-trip time is then displayed as feedback.

Protocol Used

Ping relies on the Internet Control Message Protocol (ICMP). ICMP is designed for error reporting and network diagnostics. Unlike other protocols, it does not transfer user data but instead helps identify and troubleshoot connectivity issues.

OSI Layer

ICMP operates at the Network Layer (Layer 3) of the OSI model. It works alongside the Internet Protocol (IP). In practice, ICMP messages are encapsulated inside IP packets, allowing them to travel across networks in the same way as other IP traffic.

Q2 – Ping Results with 1-Byte Packets

Command Used (Windows 11 via command Prompt):

ping www.ufl.edu -n 20 -l 1

```
C:\Users\Jalen>ping www.ufl.edu -n 20 -l 1

Pinging virtual-l2www-prod-ac-publicssl.server.ufl.edu [128.227.36.35] with 1 bytes of data:
Reply from 128.227.36.35: bytes=1 time=61ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=66ms TTL=239
Reply from 128.227.36.35: bytes=1 time=61ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=64ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=72ms TTL=239
Reply from 128.227.36.35: bytes=1 time=64ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=67ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=64ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=61ms TTL=239

Ping statistics for 128.227.36.35:
    Packets: Sent = 20, Received = 20, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 61ms, Maximum = 72ms, Average = 63ms
```

Figure 1: Ping output for www.ufl.edu with 1-byte packet)

ping www.uq.edu.au -n 20 -l 1

```
C:\Users\Jalen>ping www.ufl.edu -n 20 -l 1

Pinging virtual-l2www-prod-ac-publicssl.server.ufl.edu [128.227.36.35] with 1 bytes of data:
Reply from 128.227.36.35: bytes=1 time=61ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=66ms TTL=239
Reply from 128.227.36.35: bytes=1 time=61ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=64ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=72ms TTL=239
Reply from 128.227.36.35: bytes=1 time=64ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=67ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=65ms TTL=239
Reply from 128.227.36.35: bytes=1 time=64ms TTL=239
Reply from 128.227.36.35: bytes=1 time=62ms TTL=239
Reply from 128.227.36.35: bytes=1 time=61ms TTL=239

Ping statistics for 128.227.36.35:
    Packets: Sent = 20, Received = 20, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 61ms, Maximum = 72ms, Average = 63ms
```

Figure 2: Ping output for www.uq.edu.au with 1-byte packets

Results:

Host	**Packets Sent**	**Packets Received**	**Success %**	**Min RTT (ms)**	**Avg RTT (ms)**	**Max RTT (ms)**
www.ufl.edu	20	20	100%	61	63	72
www.uq.edu.au	20	20	100%	361	364	372

Analysis

The results show that **www.ufl.edu** (University of Florida) had much lower round-trip times, averaging about 63 ms, compared to **www.uq.edu.au** (University of Queensland), which had much higher round-trip times, averaging about 364 ms.

This difference occurs because of geographical distance and the path that network traffic takes. The University of Florida's servers are closer, so data travels a shorter distance and returns more quickly. In contrast, data sent to Australia must cross multiple international networks and cover much greater physical distances, which increases the delay.

Both hosts showed a 100% success rate, meaning all packets were received and there was no packet loss. This indicates stable and reliable connectivity to both destinations.

Q3 – Ping Results with 9000-Byte Packets

Command Used (Windows 11 via command Prompt):

ping www.ufl.edu -n 20 -l 9000

```
C:\Users\Jalen>ping www.ufl.edu -n 20 -l 9000

Pinging virtual-l2www-prod-ac-publicssl.server.ufl.edu [128.227.36.35] with 9000 bytes of data:
Request timed out.
Reply from 128.227.36.35: bytes=9000 time=85ms TTL=240
Reply from 128.227.36.35: bytes=9000 time=67ms TTL=240
Request timed out.
Reply from 128.227.36.35: bytes=9000 time=95ms TTL=240
Reply from 128.227.36.35: bytes=9000 time=79ms TTL=240
Reply from 128.227.36.35: bytes=9000 time=80ms TTL=240
Request timed out.
Request timed out.
Reply from 128.227.36.35: bytes=9000 time=80ms TTL=240
Request timed out.
Reply from 128.227.36.35: bytes=9000 time=80ms TTL=240
Reply from 128.227.36.35: bytes=9000 time=80ms TTL=240
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 128.227.36.35: bytes=9000 time=80ms TTL=240

Ping statistics for 128.227.36.35:
    Packets: Sent = 20, Received = 9, Lost = 11 (55% loss),
Approximate round trip times in milli-seconds:
    Minimum = 67ms, Maximum = 95ms, Average = 80ms
```

(Figure 3: Ping output for www.ufl.edu with 9000-byte packets)

Results:

Host	**Packet Size (bytes)**	**Packets Sent**	**Packets Received**	**Success %**	**Min RTT (ms)**	**Avg RTT (ms)**	**Max RTT (ms)**
www.ufl.edu	9000	20	9	45%	67	80	95

Analysis

The attempt I made to ping **www.ufl.edu** using 9000-byte packets was only partially successful. Out of twenty packets sent, only nine were returned, giving a success rate of 45%. Round-trip times ranged from 67 ms (minimum) to 95 ms (maximum), with most responses clustering around 80 ms.

In comparison, smaller packets had an average round-trip time of about 63 ms. The increase in time for larger packets is expected, since transmitting more data takes longer and increases the likelihood of transmission issues.

More than half of the large packets (55%) were lost in transit. This indicates that the network path between the client and the University of Florida's server does not handle oversized packets reliably. Most modern internet connections are optimized for a maximum transmission unit (MTU) of around 1500 bytes. Sending packets as large as 9000 bytes often exceeds this capacity, which leads to fragmentation or packet loss.

Q4 – How Traceroute Works

Traceroute Overview

To observe how data travels across networks, the traceroute utility (or tracert on Windows) can be used. Traceroute maps the path taken by packets from the source machine to the destination host. It does this by sending small packets with gradually increasing “time-to-live” (TTL) values.

Each router along the path decreases the TTL by one. When the TTL reaches zero, the router discards the packet and sends back a “Time Exceeded” message. Traceroute records these messages, allowing the user to identify each hop along the route and measure the round-trip time to that hop.

Why Three Times per Hop

Traceroute typically sends **three packets** for every hop. These three round-trip times show the variability of delay at each step. Having multiple samples instead of just one makes it easier to detect:

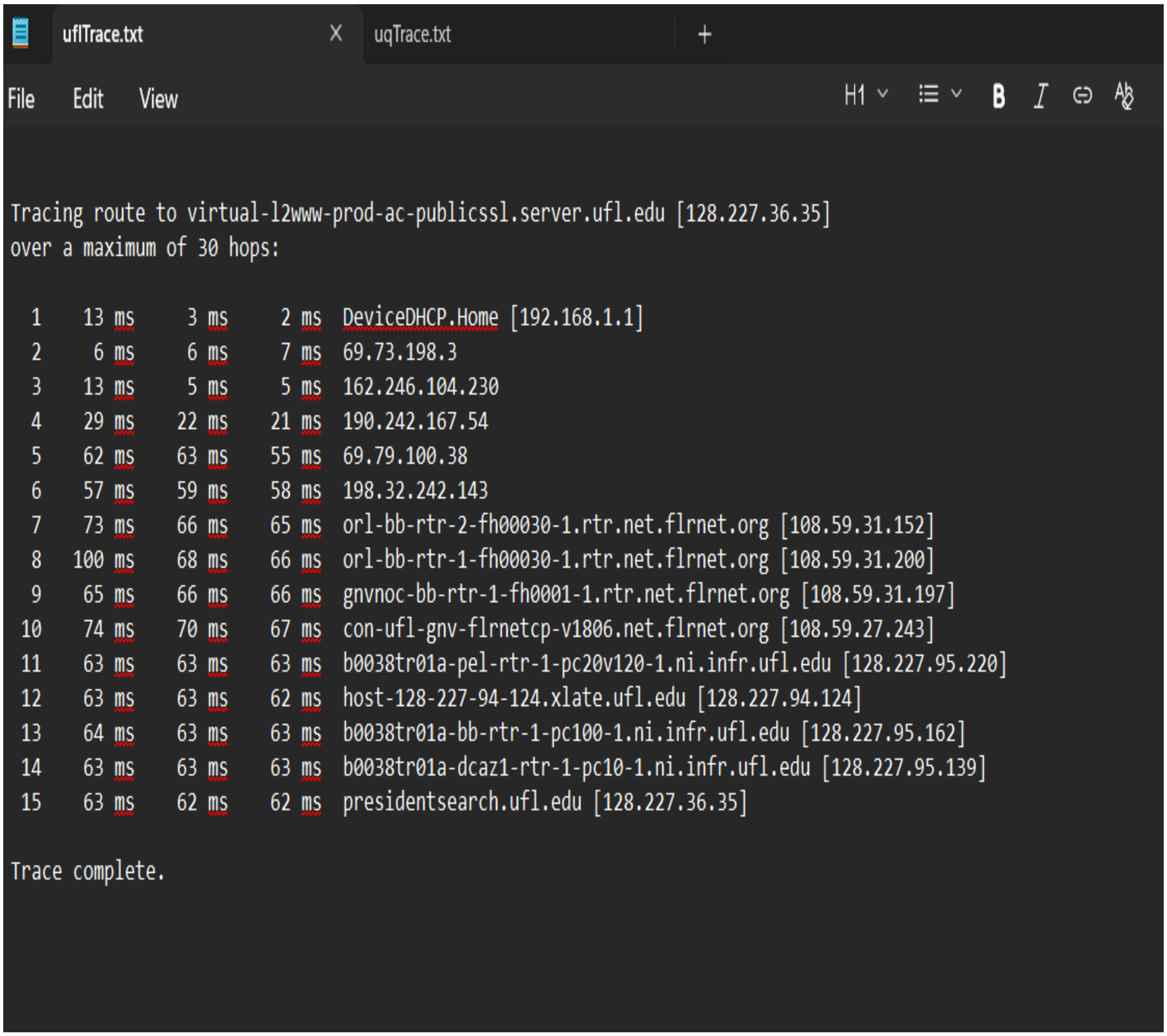
- temporary congestion,
- route changes, or
- small bursts of delay along the path.

This provides a clearer and more reliable picture of the network’s performance than a single measurement would.

Q5 – Traceroute Results for UFL and UQ

Command Used (Windows 11 via command Prompt):

- tracert www.ufl.edu > uflTrace.txt



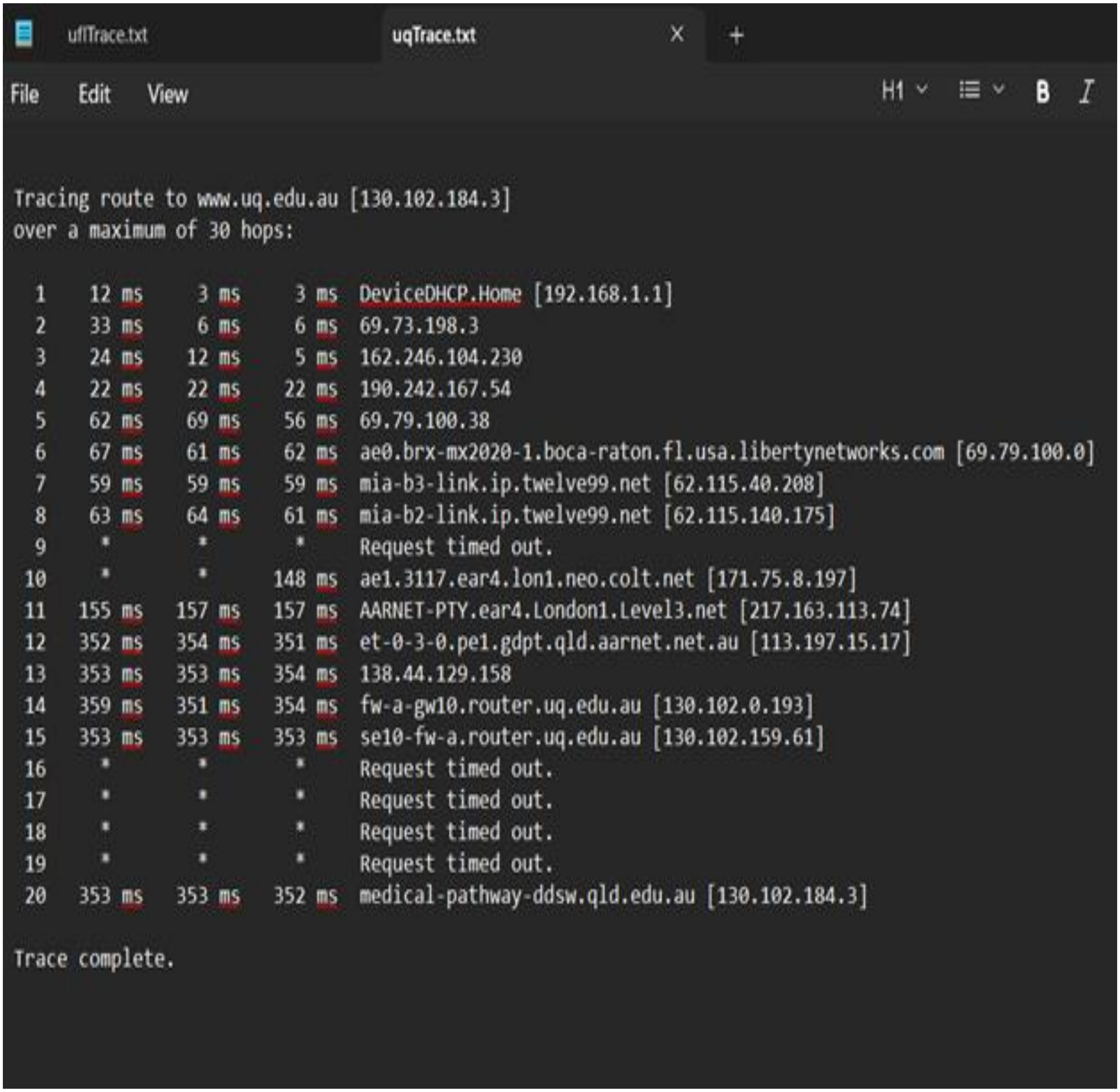

```
Tracing route to virtual-l2www-prod-ac-publicssl.server.ufl.edu [128.227.36.35]
over a maximum of 30 hops:

  1    13 ms     3 ms     2 ms  DeviceDHCP.Home [192.168.1.1]
  2     6 ms     6 ms     7 ms  69.73.198.3
  3    13 ms     5 ms     5 ms  162.246.104.230
  4    29 ms    22 ms    21 ms  190.242.167.54
  5    62 ms    63 ms    55 ms  69.79.100.38
  6    57 ms    59 ms    58 ms  198.32.242.143
  7    73 ms    66 ms    65 ms  orl-bb-rtr-2-fh00030-1.rtr.net.flrnet.org [108.59.31.152]
  8   100 ms    68 ms    66 ms  orl-bb-rtr-1-fh00030-1.rtr.net.flrnet.org [108.59.31.200]
  9    65 ms    66 ms    66 ms  gnvnoc-bb-rtr-1-fh0001-1.rtr.net.flrnet.org [108.59.31.197]
 10    74 ms    70 ms    67 ms  con-ufl-gnv-flrnetcp-v1806.net.flrnet.org [108.59.27.243]
 11    63 ms    63 ms    63 ms  b0038tr01a-pel-rtr-1-pc20v120-1.ni.infr.ufl.edu [128.227.95.220]
 12    63 ms    63 ms    62 ms  host-128-227-94-124.xlate.ufl.edu [128.227.94.124]
 13    64 ms    63 ms    63 ms  b0038tr01a-bb-rtr-1-pc100-1.ni.infr.ufl.edu [128.227.95.162]
 14    63 ms    63 ms    63 ms  b0038tr01a-dcaz1-rtr-1-pc10-1.ni.infr.ufl.edu [128.227.95.139]
 15    63 ms    62 ms    62 ms  presidentsearch.ufl.edu [128.227.36.35]

Trace complete.
```

(Figure 4: Traceroute to www.ufl.edu)

- tracert www.uq.edu.au > uqTrace.txt



```
Tracing route to www.uq.edu.au [130.102.184.3]
over a maximum of 30 hops:

  1    12 ms     3 ms     3 ms  DeviceDHCP.Home [192.168.1.1]
  2    33 ms     6 ms     6 ms  69.73.198.3
  3    24 ms    12 ms     5 ms  162.246.104.230
  4    22 ms    22 ms    22 ms  190.242.167.54
  5    62 ms    69 ms    56 ms  69.79.100.38
  6    67 ms    61 ms    62 ms  ae0.brx-mx2020-1.boca-raton.fl.usa.libertynetworks.com [69.79.100.0]
  7    59 ms    59 ms    59 ms  mia-b3-link.ip.twelve99.net [62.115.40.208]
  8    63 ms    64 ms    61 ms  mia-b2-link.ip.twelve99.net [62.115.140.175]
  9     *        *        *     Request timed out.
 10     *        *      148 ms  ae1.3117.ear4.lon1.neo.colt.net [171.75.8.197]
 11   155 ms   157 ms   157 ms  AARNET-PTY.ear4.London1.Level3.net [217.163.113.74]
 12   352 ms   354 ms   351 ms  et-0-3-0.pe1.gdpt.qld.aarnet.net.au [113.197.15.17]
 13   353 ms   353 ms   354 ms  138.44.129.158
 14   359 ms   351 ms   354 ms  fw-a-gw10.router.uq.edu.au [130.102.0.193]
 15   353 ms   353 ms   353 ms  se10-fw-a.router.uq.edu.au [130.102.159.61]
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20   353 ms   353 ms   352 ms  medical-pathway-ddsw.qld.edu.au [130.102.184.3]

Trace complete.
```

(Figure 5: Traceroute to www.uq.edu.au)

Summary Table:

Host	Reached?	Hop Count	Destination RTTs (ms)
www.ufl.edu	Yes	15	63 / 62 / 62
www.uq.edu.au	Yes	20	353 / 353 / 352

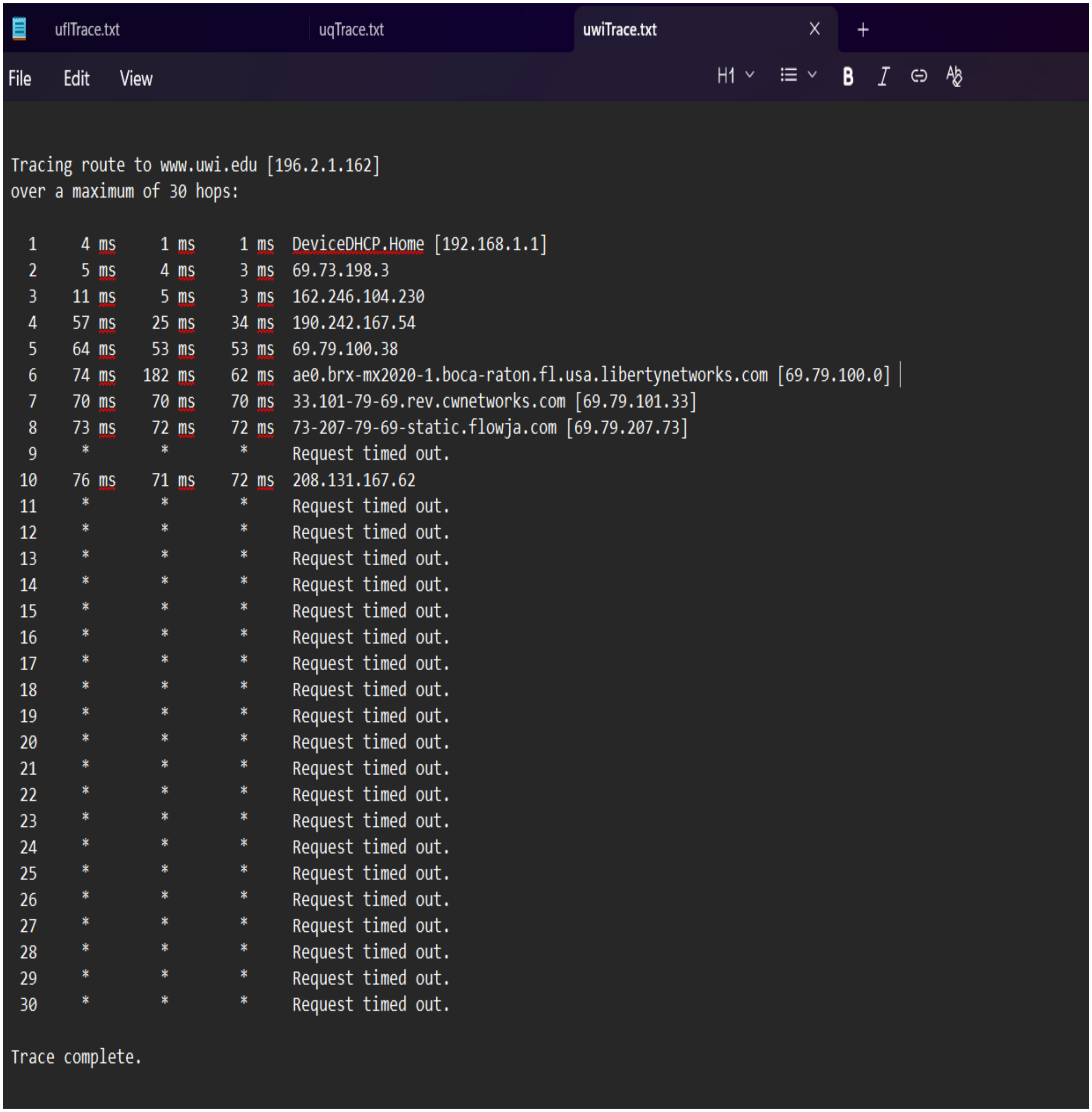
Analysis

UFL completed in **15 hops** with much lower destination RTTs than UQ.
UQ completed in **20 hops** with higher RTTs, consistent with longer distance and a longer path.\

Q6 – Traceroute Results for UWI

Command Used (Windows 11 via command Prompt):

- tracert www.uwi.edu > uwiTrace.txt




```
Tracing route to www.uwi.edu [196.2.1.162]
over a maximum of 30 hops:

  1     4 ms     1 ms     1 ms  DeviceDHCP.Home [192.168.1.1]
  2     5 ms     4 ms     3 ms  69.73.198.3
  3    11 ms     5 ms     3 ms  162.246.104.230
  4    57 ms    25 ms    34 ms  190.242.167.54
  5    64 ms    53 ms    53 ms  69.79.100.38
  6    74 ms   182 ms    62 ms  ae0.brx-mx2020-1.boca-raton.fl.usa.libertynetworks.com [69.79.100.0]
  7    70 ms    70 ms    70 ms  33.101-79-69.rev.cwnetworks.com [69.79.101.33]
  8    73 ms    72 ms    72 ms  73-207-79-69-static.flowja.com [69.79.207.73]
  9     *        *        *     Request timed out.
 10    76 ms    71 ms    72 ms  208.131.167.62
 11     *        *        *     Request timed out.
 12     *        *        *     Request timed out.
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19     *        *        *     Request timed out.
 20     *        *        *     Request timed out.
 21     *        *        *     Request timed out.
 22     *        *        *     Request timed out.
 23     *        *        *     Request timed out.
 24     *        *        *     Request timed out.
 25     *        *        *     Request timed out.
 26     *        *        *     Request timed out.
 27     *        *        *     Request timed out.
 28     *        *        *     Request timed out.
 29     *        *        *     Request timed out.
 30     *        *        *     Request timed out.

Trace complete.
```

(Figure 6: Traceroute to www.uwi.edu)

Summary Table:

Host	Reached?	Hop Count (last responder)	Destination RTTs (ms)
www.uwi.edu	**No**	**10**	**N/A**

Observation: Last responding hop was **#10** with RTTs **76 / 71 / 72 ms**. From hop **11** → **30** there were timeouts.

Analysis

Result: Destination not reached.

Two reasons:

1. **Host blocks ICMP/traceroute: www.uwi.edu or an upstream network may drop ICMP Time-Exceeded or Echo replies, so probes time out even if the site is reachable via normal web traffic.**
2. **Firewall/security filtering: Enterprise firewalls frequently block replies when their time-to-live has run out, likewise curtailing ICMP traffic — resulting in incomplete path tracing along numerous network segments and ultimately impacting visibility at the intended receiver.**

Conclusion

The lab checked connections to three university servers using Ping and Traceroute, measuring delays as well as routes taken. Smaller packets, just one byte, showed quick times at UFL (~63 ms) but were much slower reaching UQ (~364 ms); greater distances clearly meant longer travel. However, larger packets (9000 bytes) sent to UFL had problems: only about half were returned, and response times climbed to roughly 80 ms (ranging between 67–95 ms). This illustrates how large data chunks slow transmission because they encounter network size restrictions, causing breaks or lost pieces.

The network route map confirmed these observations. Reaching UFL took only fifteen steps, with response times around 63 ms each way. In contrast, reaching UQ involved twenty steps with a much slower 353 ms delay, which is expected for a connection spanning continents. Attempts to contact UWI failed; the signal stopped after ten hops, likely due to security measures that block diagnostic probes but do not affect normal website access.

Overall, latency depends heavily on distance traveled; more hops usually mean slower speeds. Packet size also matters, especially where network limits exist, since smaller packets are more reliable for delivery. Security settings can obscure route details, but they do not necessarily indicate connectivity problems. Using both Ping and Traceroute provides a clear picture of connection reliability, speed, and routing across live networks.